

Satya Sudheer Yalamarthy

Engineering leader building distributed, multi-tenant platforms at scale

Engineering Leader · 18+ years

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SUMMARY

Engineering leader with 18+ years building and operating large-scale distributed systems — multi-tenant control planes, sharded and partitioned data platforms, and real-time streaming architectures — across fraud and threat intelligence (TransUnion, ~1,000 QPS, 99.9%+ availability across regions), payments (Paytm, 12B+ daily events, 4x throughput via a partitioning-led migration), and regulated data domains. Have led engineering organizations of 20-26+ engineers directly and cross-functional delivery spanning ~25 engineers across cities via managers, setting the technical bar and operating cadence across data engineering, SRE, and product. Direct, hands-on experience building agentic RAG systems designed for the read-volume, concurrency, and authorization implications of AI-native workloads. Multi-cloud depth across AWS, GCP, and Azure.

CORE COMPETENCIES

AI / GenAI: LLMs, Retrieval-Augmented Generation (RAG), Anthropic SDK, pgvector, LangChain, agent workflows, human-in-the-loop systems, prompt engineering

Distributed Systems: Sharding & partitioning, Data migration at scale, Request routing, Multi-tenant control planes, Incident command, Capacity planning, Blast-radius containment

Frontend, Backend & Data: Python, Node.js, Scala, .NET / C#, Kafka, Apache Flink, Apache Spark, Apache Hudi, TimescaleDB, PostgreSQL, MongoDB, Cassandra, Redis, Elasticsearch, Java, Spring, Spring Boot, React, React Native, Next.js, TypeScript, JavaScript, HTML / CSS, D3 (data visualization), responsive web & mobile, design-engineering collaboration, accessibility-conscious development

Cloud & DevOps : AWS, GCP, Azure, Kubernetes, Docker, Airflow, GitHub Actions, Jenkins, Prometheus, Grafana, Datadog, Dynatrace, ELK Stack

Leadership & Process: Agile delivery, hiring & mentoring, cross-functional partnership, RFP & business development, capacity planning, vendor management, 24x7 operations

PROFESSIONAL EXPERIENCE

Senior Architect / Sr. Engineering Manager · TransUnion

Bengaluru, India · Dec 2024 – Present

- Architected the next-generation IP Intelligence (IPI) platform — a multi-tenant, multi-region control plane for real-time IP geolocation, threat scoring, and network reputation — sustaining ~**1,000 QPS** at **99.9%+** availability with deliberate blast-radius containment across regions.
- Redesigned IP, Device, and Identity Reputation services around event-driven microservice boundaries on Java, Spring Boot, Kafka, and Cassandra, migrating from batch enrichment to real-time streaming (Apache Flink) — a partitioning and data-locality redesign for multi-region scale.
- Designed and built an agentic Retrieval-Augmented Generation (RAG) system — a six-stage pipeline (intake → embed → retrieve → draft → review → publish) using a pgvector store and the Anthropic SDK behind a human-in-the-loop approval workflow — designing explicitly for the read-volume and authorization implications of agentic access rather than retrofitting them.
- Reduced p95 latency on critical reputation endpoints by ~**30%** through targeted Redis caching, query-plan optimization, and Apache Flink stream enrichment — informed by incident command and capacity-planning work across global regions.
- Led the engineering organization across data engineering, SRE, and product: mentored and developed a team of ~**20–26** engineers, instituted architecture review forums, and set the technical bar and operating cadence for cross-functional delivery.

Senior Engineering Manager · Paytm

Bengaluru, India · Sep 2022 – Nov 2024

- Architected **Pulse**, a high-volume clickstream analytics platform processing **12B+ daily events** on Kafka, Apache Flink, Apache Druid, and TimescaleDB, with self-serve DIY tooling that put real-time analytics directly in business users' hands.
- Led the settlement-processing migration: re-partitioned microservice and database boundaries on Java and Spring Boot, executing a phased, verified-rollback migration that delivered a **4x increase in daily transaction throughput** and a **29–30% infrastructure cost reduction** — cost per unit of capacity trending down as the footprint grew.

- Shipped real-time analytics capabilities across three feature groups (~25 **engineers** across multiple cities, via engineering managers), cutting analytics ingestion latency from minutes to sub-second on Apache Druid, Hudi, and TimescaleDB.
- Instituted code-review, testing, and deployment standards that lifted release frequency by ~50% and reduced production incidents quarter-over-quarter — the operating cadence for a distributed, multi-team engineering organization.
- Partnered with sales and product on enterprise RFP responses, demos, and pitches, contributing to multi-crore INR deal closures.

Associate Engineering Manager · Condé Nast Technology Labs

Bengaluru, India · Aug 2021 – Sep 2022

- Shipped the Preferences App powering newsletter personalization for Vogue, The New Yorker, GQ, Wired, Architectural Digest, and 4 other global brands, reaching millions of subscribers.
- Migrated newsletter orchestration to a React + Node.js stack on Kubernetes, improving send reliability and lifting open rates by ~30%.
- Mentored a cross-functional team of ~8 engineers and designers, establishing CI/CD on GitHub Actions, Jenkins, and Datadog observability.

EARLIER CAREER

Principal Engineer · Dun & Bradstreet, Bengaluru, India (Dec 2020 – Aug 2021)

Associate Manager · Allstate Insurance, Bengaluru, India (Jun 2014 – Dec 2020)

Senior Software Engineer · CSC (DXC), Hyderabad, India (Sep 2013 – Jun 2014)

Senior Software Engineer · L&T Infotech (LTI Mindtree), Chennai, India (Mar 2010 – Sep 2013)

Software Engineer · WinIT, Hyderabad, India (Sep 2007 – Sep 2009)

NOTABLE PROJECTS

[TransUnion] RAG-Based Jira Ticket Resolution Agent

Designing and building an AI agent that ingests customer-support tickets from Jira, retrieves similar resolved cases from a pgvector store, and proposes resolutions via the Anthropic SDK behind a human-in-the-loop approval workflow. Six-stage pipeline (intake → embed → retrieve → draft → review → publish) with evaluation hooks; four-week MVP plan with post-MVP deferrals for autonomy expansion.

[TransUnion] IP Intelligence Platform Modernisation

Architected migration of IPI reputation services to a Kafka + Flink streaming foundation on AWS and GCP, replacing batch enrichment with real-time signal computation for fraud, threat, and identity use cases.

[Paytm] Pulse 12B+ Events/Day Clickstream Analytics

Architected the end-to-end platform: Kafka ingestion, Flink stream processing, Druid + TimescaleDB serving layer, and a DIY exploration UI that put real-time business analytics directly in the hands of non-engineering teams. Became a foundational data surface for merchant- and consumer-facing decisions.

[Paytm] Settlement-Processing Refactor

Re-architected microservice boundaries on the settlement system, delivering a 4× throughput increase on daily transactions and 29–30% infrastructure cost reduction through workflow simplification, database optimization, and capacity right-sizing — while holding latency and reliability SLOs.

[Allstate] QuickFoto Claim with Image Analytics

Shipped a mobile-first first-notice-of-loss flow where customers submit photos of vehicle damage and receive estimated payouts in minutes instead of days. Combined React Native, image analytics, and event-driven payment orchestration to compress a 5-day workflow into seconds.

EDUCATION

JNTU, Hyderabad · B.Tech, Electronics and Communications Engineering

Jun 2002 – Mar 2006

CERTIFICATIONS & TRAINING

Cloud

AWS Certified Solutions Architect – Associate (Amazon Web Services, Dec 2019) · AWS Certified Developer – Associate (Amazon Web Services, May 2020) · AWS Certified SysOps Administrator – Associate (Amazon Web Services, Jul 2020) · Microsoft Certified: Azure Fundamentals (Microsoft, Oct 2020) · Google Professional Cloud Architect Certification (Google, Dec 2025)

Infrastructure & Platform

KCNA: Kubernetes and Cloud Native Associate (The Linux Foundation, Feb 2022) · HashiCorp Certified: Terraform Associate (HashiCorp, Oct 2020) · HashiCorp Certified: Vault Associate (HashiCorp, Nov 2020) · HashiCorp Certified: Consul Associate (HashiCorp, Nov 2020)

Agile

Certified SAFe® 4 Product Owner/Product Manager (Scaled Agile Inc, Nov 2019)